

OMAR ALHALAWANI

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EDUCATION

Bachelor of Engineering in Computer Systems Engineering

Ottawa, ON

Carleton University — CGPA: 10.80/12

2024 - 2029

– Ontario Professional Engineers Scholarship — Dean's Honour List

TECHNICAL SKILLS

RTL & Digital Design: SystemVerilog, Verilog, synchronous design, FSMs, datapaths, timing-aware design

Verification: SystemVerilog UVM, PyUVM, cocotb, fault injection, regression testing, waveform analysis

Programming & Automation: Python, asyncio, Bash, Makefiles, C, Modern C++, Java OOP, EDA flow automation

Tools & Workflow: Linux, Git, Vivado, simulation debugging, AI-assisted development workflows

EXPERIENCE

ASIC Engineer I Co-op – Digital Verification Methodology

May 2026 – Present

Ciena

Ottawa, ON

- Optimized Makefile-based ASIC verification compile flows with Python automation to reduce unnecessary recompilation and improve iteration efficiency.
- Evaluated PyUVM and cocotb for AI-assisted verification, generating self-checking testbenches from RTL specifications and verification plans.
- Designed fault-injection and performance experiments and debugged simulation, regression, and waveform results across verification flows.
- Documented research findings and presented verification methodology recommendations to Ciena engineers and management.

Embedded Hardware & Research Instrumentation Assistant

Sept. 2025 – April 2026

Carleton University – Engineering Laboratories

Ottawa, ON

- Developed firmware and automation scripts for real-time experimental data acquisition systems.
- Interfaced sensors, ADCs, and microcontrollers and debugged timing, signal integrity, and integration issues.
- Collaborated with researchers to validate the correctness and reliability of instrumentation subsystems.

Teaching Assistant – Programming and Digital Systems

Sept. 2025 – May 2026

Carleton University – Department of Systems & Computer Engineering

Ottawa, ON

- Mentored students in synchronous RTL design, FSM implementation, and timing-aware digital systems.
- Assisted with simulation debugging, waveform analysis, and validation of HDL-based projects.
- Reinforced structured verification practices and incremental hardware validation methodologies.

PROJECTS

Synchronous FIFO RTL Design & Verification

SystemVerilog

- Designed a synthesizable FIFO with parameterized depth and data width using synchronous design principles.
- Implemented pointer logic, full/empty status detection, and safe data handshaking.
- Developed a self-checking testbench with directed and corner-case tests and automated result validation.

FPGA-Based Arithmetic System with FSM Control

Verilog, Vivado

- Designed a structured datapath and FSM-based control unit to perform arithmetic operations on FPGA.
- Separated control and datapath logic for clarity, scalability, and timing reliability.
- Verified functionality via simulation, waveform analysis, and hardware bring-up validation.

Automated EMG Signal Acquisition & Processing Pipeline

Arduino, Python

- Implemented synchronized sampling firmware for EMG data acquisition using microcontrollers.
- Developed Python scripts to automate data logging, preprocessing, and structured output generation.
- Built repeatable analysis workflows to visualize and validate signal integrity across trials.

EXTRACURRICULAR INVOLVEMENT

Biomedical Carleton Applied Research and Engineering (BioCARE)

Sept 2025 – Present

- Developed an EMG data acquisition pipeline using MyoWare sensors and Arduino in C++.
- Streamed real-time data over serial to a Python application running on Linux.
- Built a Python GUI with live plotting and digital filtering to reduce signal noise.